Subfield parabolics and affine overgroups

Human Compression Project

Abstract

The stabilizer of a subfield line determines its projective overgroup, while its affine overgroups are controlled by invariant vector spaces. In even characteristic, one-dimensional quotients supply the additional geometries omitted by the source. For an odd extension in odd characteristic the available scalar group is too small to do so. The quadratic projective case has exactly two conjugacy classes of affine configurations, interchanged by an outer automorphism. We give these arguments in common coordinates and then treat the small exceptions separately.

1 Premises and the representation interface

Let $K = \mathbb{F}_s \subset F = \mathbb{F}_q$ and let $G = \mathrm{PSL}(2, F)$ act on $F \cup \{\infty\}$. For an additive subgroup $V \leq F$, write $U_V = \{x \mapsto x + v : v \in V\}$. Let C_D denote multiplication by elements of a subgroup $D \leq F^*$. The scalars present in the stabilizer of ∞ in G form

$$A_F = \begin{cases} F^*, & \mathrm{char}\, F = 2, \\ (F^*)^2, & \mathrm{char}\, F \text{ odd}. \end{cases}$$

All products $U_V : C_D$ below use these literal transformations.

We retain the classical Dickson subgroup classification, its subfield embedding and conjugacy statements, and its small exceptional group normalizers. In particular, odd-degree $\mathrm{PSL}(2, K)$ subgroups form one conjugacy class, as do the even-characteristic subfield groups. The two classes of $\mathrm{PGL}(2, K)$ in $\mathrm{PSL}(2, \mathbb{F}_{s^2})$ are interchanged by $\mathrm{PGL}(2, \mathbb{F}_{s^2})$. We do not infer configuration counts from these subgroup counts alone.

The following is the explicit interface with the separate common representation proof. It is not an import of the source's unpublished Proposition 4.6: for a subgroup of projective type, a two-transitive coset action has its natural Borel stabilizer, or one of the following exceptional stabilizers:

group	additional point stabilizer
$\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5)$	D_{10} or A_4, as applicable
$\mathrm{PSL}(2,7)$	S_4
$\mathrm{PSL}(2,9),\ \mathrm{PSL}(2,11)$	A_5
$\mathrm{PGL}(2,t)$ (t odd)	$\mathrm{PSL}(2,t)$ (degree two)
$\mathrm{PGL}(2,5)$	S_4

The familiar $\mathrm{PSL}(2,2) = S_3$, $\mathrm{PSL}(2,3) = A_4$ and $\mathrm{PGL}(2,3) = S_4$ actions are included with their elementary kernels. The same common proof supplies the small exceptional overgroup lists when the point stabilizer is A_4, S_4, A_5 or a small dihedral group. This interface is proved in the separately reviewed manuscript *Two-transitive coset actions in dimension two.* The coordinate and uniqueness lemmas below are independent of it.

2 A fixed Borel has a unique projective overgroup

Lemma 1. *Suppose $|K| = s > 2$ and $D \leq K^*$ is nontrivial. For $B = U_K : C_D \leq G$,*

$$\mathrm{N}_G(B) = U_K : C_{K^* \cap A_F}.$$

In particular $\mathrm{N}_G(B) = B$ in each of these cases:

1. *$s \geq 4$ is even, $D = K^*$;*
2. *$s \geq 5$ is odd, $[F : K]$ is odd, $D = (K^*)^2$;*
3. *$s \geq 3$ is odd, $[F : K] = 2$, $D = K^*$.*

Proof. The unique Sylow subgroup U_K for the characteristic of F is characteristic in B. It has the unique common fixed point ∞, so a normalizer of B has the form $x \mapsto ax + b$ with $a \in A_F$. Normalizing U_K gives $aK = K$, hence $a \in K^*$. Conjugating a scalar $c \neq 1$ in D gives translation term $(1 - c)b$. This belongs to K exactly when $b \in K$. Conversely these conditions plainly normalize B, proving the formula. For an odd-degree extension, an element of K^* is a square in F exactly when it is a square in K; for a quadratic extension every element of K^* is a square in F. The assertions follow. □

Lemma 2. *In the three cases of Lemma 1, respectively put $M = \mathrm{PSL}(2, K)$, $M = \mathrm{PSL}(2, K)$ and $M = \mathrm{PGL}(2, K)$. Assume the extension degree is prime, so that the specified M is maximal in G. Then exactly one conjugate subfield group of this type contains the fixed B as a natural Borel.*

Proof. The Borel subgroups inside M form one M-class and are self-normalizing in M. For any ambient conjugacy class of copies of M that contains B, their set above B is a transitive $\mathrm{N}_G(B)$-set: adjust a transporter inside the target M using its internal Borel conjugacy. Its stabilizer is B, because M is maximal and self-normalizing. Lemma 1 makes this set a singleton.

In the first two cases there is only one ambient class of M. In the third case there are two. The Borel subgroups have two ambient classes as well, which can be seen directly. After moving their unique fixed point to ∞, their radicals are the subfield lines aK. Conjugation multiplies a by an element of A_F. As $K^* \subset A_F$, the quotient $F^*/(A_F K^*)$ has order two. The two classes are interchanged by any nonsquare scalar in $\mathrm{PGL}(2, F)$. That same outer coset interchanges the two classes of M. Since one class of M contains only one class of Borels, the two classes of M must contain different Borel classes. Thus only one class of M can contain the fixed B, completing the proof. □

Here the radical description is literal: its scalar complement K^* forces an additive radical of order s to be a one-dimensional K-space. Any translated scalar complement can be made linear by an affine translation. The same argument gives one Borel class in the first two cases: $A_F K^* = F^*$ for an odd-degree odd-characteristic extension, and $A_F = F^*$ in even characteristic.

3 All affine covers of the fixed Borel

Lemma 3. *In any of the three cases above, let H be an affine subgroup of G containing B properly, with B maximal in H. If $D = K^*$, then the two-transitive possibilities are exactly*

$$H = U_V : C_{K^*}, \qquad K \subset V \subset F, \qquad \dim_K V = 2.$$

If $D = (K^*)^2$ and $s \geq 5$ is odd in an odd-degree extension, there is no such two-transitive affine overgroup.

Proof. An affine group containing U_K must fix ∞, since that is its only possible common fixed point. Write U_V for the translation kernel of H. The group D generates the field K: this is immediate for K^*; for $(K^*)^2$ its order is $(s-1)/2 > \sqrt{s}-1$, larger than the multiplicative group of any proper subfield. Thus V is a K-space containing K.

We first see that the scalar part of H can be taken literally. If $x \mapsto ax+b$ belongs to H, commute it with a fixed scalar $c \neq 1$ in D. The resulting translation is a nonzero K-multiple of b, so $b \in V$. Removing this translation gives the scalar a itself in H. Therefore $H = U_V : C_{D'}$ for some $D \leq D' \leq A_F$ stabilizing V. If $V = K$, then $D' \leq K^* \cap A_F = D$, contrary to properness. If $V > K$, the intermediate subgroup $U_V : C_D$ properly contains B, so maximality forces $D' = D$.

Identify the cosets of B in H with the additive quotient V/K. Translations act regularly and the point stabilizer induces exactly the scalar group D. Hence two-transitivity is equivalent to transitivity of D on the nonzero vectors of V/K. If $D = K^*$, this holds exactly when $\dim_K(V/K) = 1$: the orbit has size at most $s-1$, and a one-dimensional quotient has exactly $s-1$ nonzero vectors. For $D = (K^*)^2$, even a one-dimensional quotient has more nonzero vectors than $|D|$, so transitivity is impossible. □

The scalar case with $D = K^*$ is the previously accepted affine-plane construction. Its complete parametrization and Frobenius action are reused from *Scalar orbits and subfield configurations*; they are not a new acquisition here. Lemma 3 explains why no other affine radical or enlarged complement can occur.

Theorem 4. *Assume the representation interface in Section 1.*

1. *For even $s \geq 4$ and prime $[F : K] = m$, the configurations with $M = \mathrm{PSL}(2, K)$ and natural Borel intersection are precisely $\{M, U_V : C_{K^*}\}$ with $\dim_K V = 2$ and $K \subset V$. They have $(s^{m-1}-1)/(s-1)$ conjugacy classes. Their full semilinear classes are the Frobenius orbits on the same projective parameter set. At $m = 2$ both counts are one.*

2. *For odd $s \geq 5$ and odd prime extension degree, there is no configuration with $M = \mathrm{PSL}(2, K)$ and its natural Borel intersection.*

3. *For odd $s \geq 3$ and $F = \mathbb{F}_{s^2}$, there are exactly two conjugacy classes and one automorphism class of configurations with $M = \mathrm{PGL}(2, K)$ and natural Borel intersection. Their other parabolic is $U_F : C_{K^*}$.*

Proof. The representation interface and the characteristic restrict a non-affine overgroup of the given Borel to the same projective type as M. Indeed, the natural Borel stabilizer determines its field from its characteristic part and scalar order. A cyclic group cannot contain B. A dihedral candidate can occur only for the small coincidences $B = D_{10}$ in characteristic five or $B = D_6$ in characteristic three; a proper dihedral overgroup would require a cyclic element with mixed characteristic order, or order nine in characteristic three, neither of which occurs in a projective group in that characteristic. The exceptional groups A_4, S_4, A_5 are checked by their orders and the small-overgroup interface. The small coincidence $U_4 : C_3 = A_4$ gives $A_5 = \mathrm{PSL}(2, 4)$ in characteristic two; the coincidence $U_5 : C_2 = D_{10}$ gives $A_5 = \mathrm{PSL}(2, 5)$ in characteristic five. They introduce no new projective overgroup. Lemma 2 excludes a second such copy. The remaining possibilities are exactly those in Lemma 3.

For the first case, the accepted scalar-orbit proof gives the stated counts and parameter set. In the second case there is no affine possibility. In the third case the only two-dimensional K-space is F itself. Each fixed Borel therefore has exactly one projective and one affine overgroup, so

its two ambient classes give exactly two configuration classes. A nonsquare scalar exchanges the two, proving the single automorphism class. All intersections are exactly B; maximality of M gives generation of G, and the affine action on F/K supplies local two-transitivity. □

4 Odd subfields and the exceptional stabilizers

Here a subgroup defined over K means that every projective element lies in $\mathrm{PGL}(2, K)$ and is individually fixed by the Frobenius of F/K. Mere stability of the subgroup as a set is insufficient.

Lemma 5. *Let $K \subset F$ be finite fields. Suppose $L_1, L_2 \leq \mathrm{PGL}(2, K)$ have trivial centralizer in $\mathrm{PGL}(2, F)$. Every element of $\mathrm{PGL}(2, F)$ conjugating L_1 to L_2 belongs to $\mathrm{PGL}(2, K)$. If the extension degree is odd, two such subgroups of $\mathrm{PSL}(2, K)$ that are not conjugate there cannot become conjugate in $\mathrm{PSL}(2, F)$.*

Proof. Let $gL_1g^{-1} = L_2$ and let σ be the Frobenius generating $\mathrm{Gal}(F/K)$. For every $h \in L_1$, both h and ghg^{-1} are fixed by σ. Thus $g^{-1}\sigma(g)$ centralizes L_1 and is the identity. A projective matrix fixed by σ has a representative over K: normalize one nonzero entry to one, after which all entry ratios are fixed. Hence $g \in \mathrm{PGL}(2, K)$. Finally $\mathrm{PGL}(2, K) \cap \mathrm{PSL}(2, F) = \mathrm{PSL}(2, K)$ for odd extension degree, by the square criterion used in Lemma 1. □

Lemma 6. *Let M be the standard $\mathrm{PSL}(2, K)$ or $\mathrm{PGL}(2, K)$ in $\mathrm{PGL}(2, F)$. A subgroup $B \leq M$ with trivial projective centralizer is contained in no other $\mathrm{PGL}(2, F)$-conjugate of M.*

Proof. Suppose $B \leq gMg^{-1}$. Then $g^{-1}Bg$ and B are both subgroups of $\mathrm{PGL}(2, K)$, and their centralizers are trivial. Lemma 5 gives $g \in \mathrm{PGL}(2, K)$. That group normalizes both possible choices of M, so $gMg^{-1} = M$. □

In the odd-characteristic fields used in this section, the groups A_4, S_4, A_5 have trivial projective centralizer. Here is an elementary check. In characteristic three choose two distinct cyclic subgroups of order three: their unipotent centralizers have different fixed points and intersect trivially. Indeed, two distinct such subgroups with the same fixed point would commute and generate a group of order nine, which cannot lie in any of the three groups under consideration. In other characteristics the centralizer of each such subgroup is its torus. An element centralizing both must lie in both tori. A nonidentity noninvolution has its own unique torus centralizer; an involution has a dihedral centralizer whose odd-order subgroups lie in its torus. In either case both subgroups of order three would be the same unique subgroup of order three in that torus, a contradiction.

Theorem 7. *Assume the common representation interface. For $q = s^m$ odd with m an odd prime and maximal $M = \mathrm{PSL}(2, s)$, the only configurations are those with $s = 3$ and two A_4 parabolics meeting in C_3. Their counts are the corrected counts in the accepted scalar-orbit proof. In particular no additional configurations arise from the exceptional A_4, S_4, A_5 stabilizers at $s = 5, 7, 9, 11$.*

Proof. For $s \geq 5$ the natural Borel case is excluded by Theorem 4. The exceptional point stabilizer is A_4 at $s = 5$, S_4 at $s = 7$, and A_5 at $s = 9, 11$. The small overgroup interface in the actual characteristic leaves only another copy of the same M as a two-transitive overgroup. All copies of this subfield group are conjugate under $\mathrm{PGL}(2, F)$, by the retained subfield classification. The exceptional stabilizers have trivial projective centralizer, as just proved. Lemma 6 therefore gives

a unique M containing the fixed stabilizer, ruling out a second parabolic. This argument also explains why simply fusing internal subgroup classes would not be an adequate replacement for a literal overgroup proof.

At $s = 3$, $B = C_3$ is unipotent. Its normalizer in G is U_F: an affine multiplier preserving the translation line $\mathbb{F}_3$ is ± 1, and -1 is a nonsquare in F because m is odd. The elementary degree argument for a subgroup with point stabilizer C_3 gives C_6, S_3, A_4. The first two would normalize C_3 and contain an involution, impossible in U_F. Only the A_4 overgroups remain. Their complete conjugacy and Frobenius-orbit counts, with unordered interchange, were proved in the accepted scalar-orbit component. □

5 The small even-field cases

Theorem 8. *The case $M = \mathrm{PSL}(2,2)$ in $G = \mathrm{PSL}(2,4)$ gives exactly one conjugacy and one automorphism class of each of the following types:*

$$(S_3, S_3; C_2), \qquad (S_3, C_2^2; C_2), \qquad (S_3, A_4; C_3).$$

For $M = \mathrm{PSL}(2,4)$ in $G = \mathrm{PSL}(2,4^m)$ with m prime, the exceptional intersection D_{10} gives the single type $(A_5, D_{30}; D_{10})$, with one class in both senses, exactly when $15 \mid 4^m \pm 1$. This condition is equivalent here to $m = 2$.

Proof. Use the natural A_5 model for $\mathrm{PSL}(2,4)$. A subgroup S_3 is the normalizer of a three-cycle subgroup; its involutions are products of two transpositions. It is maximal in A_5. There are ten such subgroups, fifteen involutions, five point stabilizers A_4, and five subgroups C_2^2; each latter group is the centralizer of any of its nonidentity elements. These counts also follow directly by choosing the supports of cycles in five letters.

A point stabilizer C_2 in a two-transitive action has degree two or three, since $d - 1 \mid 2$. Thus its overgroup has order four or six. The order-four possibility in A_5 is C_2^2, uniquely specified by the involution. The order-six possibility is S_3. Each involution belongs to exactly two S_3 subgroups: count $10 \cdot 3 = 30$ incidences and use transitivity on the fifteen involutions. Their unordered pair is unique, proving the first configuration count. For the different-type pair S_3, C_2^2, all chains consisting of S_3 and one of its involutions are conjugate, proving the second count.

For the intersection C_3, the small degree argument leaves C_6, S_3, A_4. There is no C_6 in A_5, and the normalizer S_3 of the fixed C_3 is its unique order-six overgroup. Exactly two point stabilizers A_4 contain this C_3, corresponding to its two fixed letters. Its S_3 normalizer interchanges these two letters. Hence there is exactly one configuration of the third type. Each asserted intersection and each local action follows from the indicated elementary subgroup lattices. Maximality of the S_3 parabolic gives generation and connectedness.

Now let $q = 4^m$. The two-transitive overgroup interface for D_{10} leaves A_5, D_{20}, D_{30}. The cyclic torus containing C_5 has odd order $q \pm 1$. The normalizer of the specified D_{10} is D_{10} itself: a torus element normalizing it has square in C_5, and squaring is invertible on the odd-order quotient torus. There is one class of A_5 subfield groups and one internal class of D_{10}, so the transporter argument gives a unique A_5 above the fixed intersection. An overgroup D_{20} is impossible, since it would contain an element of order ten in characteristic two.

A D_{30} overgroup exists precisely when the same torus contains C_{15}, and is then uniquely $\langle C_{15}, D_{10} \rangle$. The D_{10} subgroups form one ambient class, proving both counts. Their local indices are six in A_5 and three in D_{30}; the corresponding natural actions are two-transitive. At $m = 2$ the condition holds because $4^2 - 1 = 15$. If m is odd, five divides $4^m + 1$, while three divides $4^m - 1$, so no torus order is divisible by fifteen. This proves the last assertion. □

6 The remaining quadratic projective intersections

Theorem 9. *Let s be odd and $G = \mathrm{PSL}(2, s^2)$, with maximal parabolic $M = \mathrm{PGL}(2, s)$. An intersection $\mathrm{PSL}(2, s)$ gives no second parabolic unless $s = 3$, in which case the other parabolic is A_5. That exceptional configuration has two conjugacy classes and one automorphism class. The exceptional S_4 intersection at $s = 5$ gives no configuration with a second parabolic.*

Proof. Put $L = \mathrm{PSL}(2, s)$. Its ambient normalizer is M, by the classical subfield normalizer formula. Any other $\mathrm{PGL}(2, s)$ overgroup would have L as its characteristic subgroup of index two, and would therefore lie in $\mathrm{N}_G(L) = M$. Hence it equals M. The representation and small-overgroup interface leaves only A_5 as an additional possibility, and only when $s = 3$, so that $L = A_4$. For clarity, an overgroup of larger projective field cannot create a hidden option: the subfield classification forces its field to contain $\mathbb{F}_s$. Inside $\mathbb{F}_{s^2}$ the only larger possibility is the entire ambient group. Its index over L is $s(s^2 + 1)$, which exceeds $|L| + 1$ and violates the two-transitivity order bound.

For $s = 3$, the group $G = A_6$ has two classes of A_4, with normalizer S_4, and two classes of A_5. Each A_5 has one internal class of A_4. An outer automorphism interchanges the two classes of each type; hence the two A_5 classes contain different A_4 classes. Fix $L = A_4$. The members of its one applicable A_5 class containing it form a transitive $\mathrm{N}_G(L) = S_4$-set with stabilizer L: the A_5 is self-normalizing and its intersection with S_4 is precisely L. Thus there are two such A_5 overgroups, exchanged by the fixed S_4 parabolic. This gives one configuration class for each of the two A_4 classes, and therefore two in all. The outer automorphism fuses them, giving one automorphism class. The local indices are two and five, with the standard two-transitive actions.

At $s = 5$, the subgroup S_4 is the exceptional point stabilizer in $M \cong S_5$. The common overgroup interface leaves only $\mathrm{PGL}(2, 5)$. Its S_4 stabilizer has trivial projective centralizer and is defined over $\mathbb{F}_5$. Lemma 6 gives a unique $\mathrm{PGL}(2, 5)$ overgroup, ruling out a second parabolic. □

The $q = 9$ configuration with two S_4 parabolics and intersection D_8 is imported from the separately accepted duad–syntheme proof, which also identifies its full graph group. It is not proved again here.

7 Source claims, corrections and attribution

The original numerical rows are interpreted under their stated maximal-parabolic hypotheses. The results above give the following joins; the common representation interface remains a separately reviewed dependency.

- L063 Lemma 5.11 and Proposition 5.12, and L064 Lemma 21 / Proposition 22: Theorem 4 recovers Table 4 row 1 at $m = 2$ and supplies the missing affine families for other prime degrees. The published restriction to $m = 2$ is false. The fixed appendix's Lemma 21 already lists the affine type without that restriction, so its statement is distinguished from the false exhaustivity conclusion of Proposition 22.

- Table 4 rows 2–4 are Theorem 8 at $q = 4$. Row 5 repeats Table 1 row 1 and is the D_{10} part of that theorem. The natural-Borel subcase at $s = 4$ also reuses the accepted Table 1 rows 2–4 scalar-orbit counts.

- L063 Lemmas 5.13–5.15 / Proposition 5.16, and L064 Lemmas 23–25 / Proposition 26: Theorem 7 proves the remaining exclusion arguments in their Section 5.4.2 context of odd

prime extension degree. The two surviving Table 5 rows retain the accepted correction for unordered inner counts. The statement of Lemma 5.13 is read with that surrounding odd-degree hypothesis.

- L063 Lemma 5.18 / Proposition 5.21 and L064 Lemma 28 / Proposition 30: Table 6 row 1 is Theorem 4(3). Theorem 9 gives row 2 and the nonexistence conclusions above $\mathrm{PSL}(2, s)$ and, at $q = 25$, above S_4. These also cover L063 Lemmas 5.19–5.20 and L064 Lemma 29. The fixed appendix's Proposition 30 does not include a separate $q = 25/S_4$ subcase; that published-source obligation is proved explicitly here.

  The proof of Lemma 24 in the fixed appendix also states that $\mathrm{PSL}(2, 3^{2m})$ contains no $\mathrm{PGL}(2, 3^w)$ subgroup. That auxiliary assertion is false: $w = 1$ and $w = m$ are allowed subfield embeddings. It is not used here. In the latter case, 5 does not divide $|\mathrm{PGL}(2, 3^m)|$ for odd m, so it cannot contain A_5; the full exclusion is supplied by the explicit representation interface.

- Table 6 row 3, L063 Lemma 5.17 and L064 Lemma 27 are joined to the accepted $q = 9$ component, together with the small overgroup list. Their full graph properties are already covered separately.

The original family constructions and classical subgroup theory retain their attribution. The coordinate affine-cover argument and the Galois-transporter descent argument are the common proofs used here; the earlier scalar-orbit and exceptional-geometry manuscripts are explicitly reused. This manuscript does not claim a new principal refutation, a new answer to an open conjecture, Lean verification or a compression measurement. Whole-original disposition requires the remaining source components and their exact claim ledgers.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Sections 5.4–5.5 and Tables 4–6.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition, Lemmas 21–29 and Propositions 22, 26, 30.